

INSOMNIA

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INTRODUCTION

Insomnia is a common complaint in the primary care setting, and primary care physicians must be adept in screening through the complaint in an efficient manner to ensure that there are no underlying mental health comorbidities that masquerade as insomnia. Many mental health issues surface as insomnia, as patients consider it more acceptable to have this complaint rather than telling physicians that they are depressed or anxious, especially in the older generation. Sleep is important for growth, learning, development, and mood.

Points About Sleep

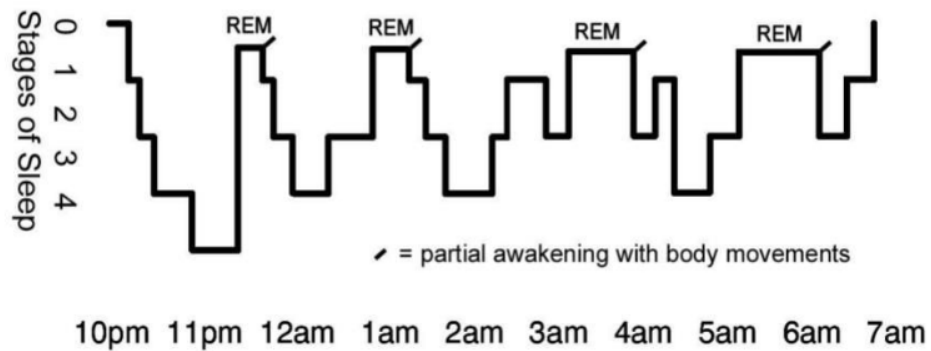
- Cycles of around 90min by 3 years old (shorter cycles in infancy) – light sleep (high levels of brain activity, dreaming, muscles paralysed to avoid acting out dream) vs deep sleep (can move around, but low brain activity)
- Circadian troughs (dips) – late afternoon, middle of the night (when you feel the most sleepy)
- Circadian peaks – early morning, evening (when children get a ‘second wind’)
- Sleep deprivation can lead to learning problems, poor attention, hyperactivity, difficulty with memory-related tasks, task organization and planning, obesity, more frequent illnesses.
- An approximate guide of how much sleep is required is important especially for infants, children, and adolescents. However, in adulthood these guides, e.g., 8 hours/day, may become a source of stress and perpetuate anxiety related to ‘not getting enough sleep’ which worsens the problem.

Newborns	14-17 hours/day
1-11 months	12-15 hours/day
1-2 years old	11 to 14 hours/day
3-5 years old	10 to 13 hours/day
6-13 years old	9 to 11 hours/day
13-17 years old	8 to 10 hours/day

Blue light (iPads / phones / tablets) blocks melatonin, so stops us from going to sleep.

Normal Stages of Sleep

Sleep is organized into cycles- 4 non-REM and one REM cycle...deepest sleep early



SCREENING AND DIAGNOSIS

What It Is (DSM-5)

1. Complaint of dissatisfaction with **sleep quality/quantity**
 - Difficulty initiating OR maintaining sleep, OR early morning awakenings
2. **Causing** significant **distress** or **impairment in functioning**
3. Occurs at least **3 nights/week**, despite **adequate opportunity** to sleep
4. Present for **at least 3 months**
5. Sleep difficulty occurs despite adequate opportunity for sleep.
6. The insomnia is not better explained by and does not occur exclusively during another sleep-wake disorder (e.g., narcolepsy, a breathing-related sleep disorder, a circadian rhythm sleep-wake disorder, a parasomnia).
7. The insomnia is not attributable to the physiological effects of a substance (e.g., a drug of abuse, a medication).
8. Coexisting mental disorders and medical conditions do not adequately explain the predominant complaint of insomnia

* *40-50% of those with insomnia have comorbid mental illness*

* *Ask for eczema, low mood, anxiety, restless legs (check ferritin levels and treat if needed), recent changes in environment, pain, drug, or substance use, shift work pattern, symptoms of obstructive sleep apnoea, nocturnal seizures, stimulant ingestion e.g., coffee/tea/nicotine/chocolates, and noisy environment*

When it is significant

When daytime somnolence occurs.

Do use the Insomnia Severity Index (ISI) - available as part of Epic Flowsheets, as part of your evaluation of the complaint if you suspect insomnia to be the primary cause of sleep issues. Interpretation of ISI is as follows:

Total score	Interpretation	Treatment / Referral recommendations
0-7	No clinically significant insomnia	Usual care
8-14	Subthreshold insomnia	TCU MSW within 8 weeks for self-help strategies, sleep hygiene, and monitoring of symptom progression
15-21	Clinical insomnia (moderate severity)	If no-low suicide risk, TCU Psychology (Short) FV for CBT-I (insomnia-specific CBT) within 4 weeks TCU Dr HMC (first Visit) for consideration of medication If moderate suicide risk, refer Psychiatry SOC (Direct access) with safety planning done (.nupsafetyplan) If high suicide risk, refer relevant emergency department
22-28	Clinical insomnia (severe)	If no-low suicide risk, TCU Psychology (Short) FV for CBT-I (insomnia-specific CBT) within 2 weeks If moderate suicide risk, refer Psychiatry SOC (Direct access) with safety planning done (.nupsafetyplan) If high suicide risk, refer relevant emergency department

For those with clinical insomnia regardless the severity, do proceed to the following interventions that could help, as below.

SPECIAL SITUATIONS

Delayed Sleep Wake Phase Disorder (DSWPD)

Happens in 3-16% of youths. Not true insomnia but causes trouble in functioning because of demands of school / societal norms. Have normal sleep latency, normal sleep maintenance and duration, if allowed to go to bed at their own schedule.

Get sleep diary of 2-week duration from patients.

Management of DSWPD	Examples
Bright light therapy	Open blinds / curtains in the room to allow natural sunlight in at the appropriate timing
Bedtime fading (when > 30min between going to bed and falling asleep)	Temporarily set bedtime later, bring forward bedtime by 15 min every 2 nights (may take 7-10 nights, like adjusting from jet lag). Avoid weekend sleep ins and naps
Chronotherapy (only works for the motivated, principle of increasing sleep drive by keeping sleep to less than the 9 hours recommended for adolescents) Delayed Sleep Wake Phase Disorder in Adolescents: Chronotherapy and Best Practices (psychiatrictimes.com)	Day 1 – sleep 3am, awake 11am Day 2 – sleep 6am, awake 2pm Day 3 – sleep 9am, awake 5pm Day 4 – sleep 12pm, awake 8pm Day 5 – sleep 3pm, awake 11pm Day 6 – sleep 6pm, awake 2am Day 7 – sleep 9pm, awake 5am Day 8 – sleep 10pm, awake 6am Need to maintain for 2 months after achieving the time even on weekends / holidays to help ‘set’ internal clock.

MANAGEMENT

Non-Pharmacological	Pharmacological
• Cognitive behavioural therapy for insomnia (CBT-I) • Sleep restriction therapy • Sleep hygiene • Stimulus control • Relaxation training	• Over the counter (OTC) • Antihistamines • Benzodiazepines • Z-drugs • Psychiatric drugs

Non-Pharmacological

Management	Examples
Cognitive Behavioural Therapy for insomnia (CBT-I) (First line therapy)	Better long-term effectiveness than medications. 4-6 sessions of therapy that includes: 1. Cognitive interventions • Cognitive restructuring: Restructuring the thoughts, feelings, and behaviours that are contributing to insomnia (e.g., challenge false belief that everyone needs 8 hours of sleep for good health) • Constructive worry time: If night-time worry is a significant issue, set a 'constructive worry' time of 15 mins to perform problem solving task using structured problem solving, at least 2 hours before bed. If one begins to worry at bedtime, remind self that the issues have been dealt with in the best way he/she knows, and he/she can continue working on them the next day. 2. Behavioural interventions • Stimulus control • Sleep Restriction • Relaxation training 3. Psychoeducation component • Sleep education to enable patient to better understand how the human biological sleep system functions • Psychoeducate on misconceptions and anxiety-provoking beliefs about sleep so that they may develop realistic sleep expectations.

Management	Examples
Sleep Restriction Therapy	Behavioural treatment that involves shortening the amount of time spent in bed to consolidate sleep. Step 1. Determine allowed time in bed (average sleep time + 30mins, no less than 5 hours) Step 2. Set a standard wake up time. Step 3. Determine bedtime by counting back from wake time the amount of allowed time in bed (referencing Step 1) Step 4. When sleep efficiency (time they spend asleep divided by the time they spend in bed as a percentage) reaches 90%, can increase time in bed by 15min every week until patient is sleeping better at night and feel good during the day
Sleep Hygiene	Avoid stimulants / caffeine / nicotine / alcohol. Exercise (at least 2 hours before bedtime). No clock watching, no electronic devices in bed, nap before 3pm if needed but not > 1 hour. Cool / dark environment
Stimulus control	Go to bed only when sleepy. Go to another room if unable to sleep within 15-20 min. Bedroom is for sleep and sex only
Relaxation training	1. Progressive muscle relaxation: Methodical tensing and relaxation of different muscle groups throughout the body 2. Diaphragmatic breathing exercise 3. Autogenic training: repeat a set of self-guided visualisations that induce a state of relaxation

Pharmacological

Drug	Usual Dose for Insomnia	Common ADR (Adverse Drug Reactions)	Remarks / Contraindications / Precautions
Melatonin			
Circadin 2mg prolonged release tablet Available in NUP (1 box has 21 tablets)	2mg 1-2 hours before bedtime, for up to 13 weeks, and can be prescribed up to 52 weeks safely. For first prescription, can prescribe 12 weeks duration, with remarks to supply 1 box first (21 tablets) and collect the rest if suitable.	Not well-established. Vivid dreams and nightmares, dizziness, daytime sleepiness, headache, irritability, and stomach cramps have been reported	For sleep onset insomnia. Consider in elderly or patients with cognitive dysfunction, glaucoma, benign prostate hyperplasia. Circadin 2 mg prolonged-release tablets are available as a Prescription-Only Medication and may be prescribed by all NUP prescribers for patients aged 55 years and above, for a maximum duration of 13 weeks.
OTC	3, 5, 10mg strengths available OTC. May start with lower doses. No evidence that increasing dose is more effective.		For off-label use in patients below 55 years of age and/or for extended treatment between 13 and 52 weeks, all prescribers may prescribe Circadin only under specific conditions: the patient must have had an unsatisfactory trial of OTC melatonin, must agree to the higher cost of Circadin, or must either be stepped down from SOC or have previously received Circadin PR 2 mg from HMC or MC doctors. For first-time prescriptions, the duration should not exceed 13 weeks, and the prescription should include a remark to dispense one box initially. For patients requiring longer-term use (13 to 52 weeks), the prescription should include the remark: “.fdate: requires longer term use.” OTC supplement is not routinely recommended as not of medicinal grade.

Drug	Usual Dose for Insomnia	Common ADR (Adverse Drug Reactions)	Remarks / Contraindications / Precautions
Orexin Antagonist Lemborexant (Dayvigo)			
Sedating Antihistamines			
Promethazine (S1) (\$0.20/tab)	12.5-25mg ON	Sedation, dizziness	Hydroxyzine may be useful for insomnia associated with anxiety.
Hydroxyzine (S1) (\$0.20/tab)	10-50mg ON	Caution in elderly, increased risk of anticholinergic side effects (e.g., dry mouth, eyes, urinary hesitancy, confusion)	Tolerance may develop with extended use.
Chlorpheniramine (S1) (\$0.08/tab)	4mg ON		
Diphenhydramine (Only URTI combination product available in NUP formulary)	25-50mg ON		
Benzodiazepines			
Alprazolam (Xanax) (NS) (\$0.20/tab)	0.25mg ON	Somnolence, drowsiness, dizziness, ataxia	Adjunctive or bridge therapy for insomnia due to anxiety or depression. Limit use to <2 weeks and take once every 2-3 nights only when necessary. Refer psychiatry if unable to discontinue BZD use. Must always document clinical review and indication in cases where BZD continues to be repeatedly prescribed. Dependence and habituation may occur. Rapid or abrupt discontinuation may precipitate withdrawal. Risk of excessive sedation, cognitive impairment, delirium and falls in elderly. Avoid in opioid/ substance use disorder. Increased risk of sedation, respiratory depression, coma, and death with concomitant opioid use.
Lorazepam (Ativan) (S1) (\$0.20/tab)	0.5mg-1mg ON		
Diazepam (Valium) (S1) (only injection and rectal solution available in NUP formulary, usually used for other indications)	Tablet: 2mg-5mg ON		
Clonazepam (S1) (\$0.37/tab)	0.5mg ON		
Z-Hypnotics			
Zopiclone (NS) (\$0.30/tab)	3.75mg-15 mg ON 3.75 mg ON in elderly		For sleep onset and sleep maintenance insomnia.

Drug	Usual Dose for Insomnia	Common ADR (Adverse Drug Reactions)	Remarks / Contraindications / Precautions
Zolpidem ER (Not available in NUP)	6.25-12.5 mg ON 6.25 mg ON in elderly	Somnolence, drowsiness, dizziness, ataxia, next-morning residual sedation Increased risk of excessive sedation, cognitive impairment, delirium and falls in elderly	Rarely, may cause complex sleep-related behaviours including sleep-talking and sleepwalking, which may lead to injury or death. Avoid in opioid or substance use disorder, or history of complex sleep-related behaviour.
Antidepressants with sedative effects			
Mirtazapine (S2) \$\$^	7.5mg-30mg ON (for sedative properties) Up to 45mg/day (for depression) Higher doses may cause excessive activation	Somnolence, increased appetite, weight gain, dizziness. Caution in those with pre-existing metabolic syndrome.	For patients with anxiety or depression. May cause cognitive and motor impairment. Risk of suicidal ideation and behaviour in children and young adults.
Fluvoxamine (S2) \$\$\$^	50-300mg/day in 2 divided doses (for depression/ anxiety)	Somnolence, nausea vomiting, diarrhoea, dizziness, nervousness, dry mouth. May cause sexual dysfunction.	May cause hyponatremia, serotonin syndrome, QT-prolongation, bleeding, activation of mania or hypomania. Avoid in angle-closure glaucoma.
Amitriptyline (S1) \$^	10mg-50mg ON Up to 300mg/day (for depression)	Somnolence, weight gain, dry mouth, constipation, dizziness, headache. Avoid in the elderly.	Caution in seizure disorders.
Trazodone (not in NUP formulary)	25-50mg ON, up to 200mg ON	Orthostatic hypotension, syncope, oedema, blurred vision, diarrhoea, nasal congestion, weight loss. May cause priapism.	

Drug	Usual Dose for Insomnia	Common ADR (Adverse Drug Reactions)	Remarks / Contraindications / Precautions
Antipsychotics with sedative effects			
Quetiapine (S2) \$^	25-100mg ON Higher doses may be used for psychosis	Sedation, nausea, vomiting, constipation, dry mouth, orthostatic hypotension, headache, weight gain	For patients with concomitant psychosis or BPSD.
Risperidone (S2) \$\$\$\$^	0.5-8mg/day (for psychosis/schizophrenia)		Second generation antipsychotics can augment anti-depressants' effect for depression, anxiety disorders, and obsessive-compulsive disorder. Consider using it for sedation and augmentation strategies.
Olanzapine (S2) \$\$\$\$^	2.5-30mg/day (for psychosis/schizophrenia)		Higher risk of weight gain, diabetes, dyslipidaemia with olanzapine. Higher risk of extrapyramidal symptoms and tardive dyskinesia with risperidone. Rarely, may cause neuroleptic malignant syndrome, seizures, agranulocytosis, and an increased risk of mortality from all causes, especially in older adult patients with dementia-related psychosis.

^ Cost \$ <\$10 per month \$\$ \$10 - <\$20 per month \$\$\$ \$20 - <\$30 per month
 \$\$\$\$ \$30 - <\$60 per month \$\$\$\$\$ \$60 - <\$90 per month \$\$\$\$\$\$ \$90 or more per month

Amount payable depends on patient subsidy level and drug subsidy class (Standard Drug List S1 and S2, Non-Standard Drug List S3 and S4)
 Unit prices before GST and accurate as of Dec 2023

Prices shown are an estimate. Please contact pharmacy for accurate pricing.

Role of Health Team Members

1. Family Physician (Dr)

- TCU NUP Psychologist for CBT-i. If medications are also required, then to refer Dr Health and Mind Clinic (HMC) in NUP (those aged ≥ 18 years with depression, anxiety, adjustment disorders, insomnia).
- Manage stable patients discharged from HMC.
- Refer to Psychiatry SOC based on geographical boundaries for those with high risk of suicide or risk of harm to self or others, or patients who are part of the exclusion list of HMC e.g., addictions / legal issues / bipolar disorder / new onset psychosis / new onset OCD.

2. Care Manager in Health and Mind Clinic (HMC CM)

- Conduct depression, anxiety, insomnia, and suicide risk screening for patients visiting HMC Dr for the first time.
- Offer psychoeducation and basic self-help techniques,
- Details to refer to [NUP-WI-CS-COP-030 Management of Patients in Health and Mind Clinic](#).

3. Family Physician in Health and Mind Clinic (HMC Dr)

- Manage new cases and follow-up cases of depression / anxiety / adjustment disorders / insomnia who are 18 years old and above.
- Follow-up psychiatric stepdown cases that are not suitable for teamlet care.
- Escalate patients to SOC if needed.
- Details to refer to [NUP-WI-CS-COP-030 Management of Patients in Health and Mind Clinic](#).

4. Psychologist

- Provides psychological assessment and intervention to individuals with mental health conditions who are part of Psychology services inclusion list.
- Conduct psychological screening and assessment during initial assessment to assess the severity of mental health symptoms, complexity of issues and presence / absence of risk tendencies and behaviours.
- Formulate treatment plan for follow-up visits.
- Provides psychological and behavioural interventions to manage mental health symptoms, improve disease management and quality of life.

5. Medical Social Worker

- Provide assistance and support for the following areas:
 - ✓ Basic sleep hygiene, self-help strategies
 - ✓ Care assessment and arrangement
 - ✓ Supportive counselling
 - ✓ Crisis intervention
 - ✓ Advance Care Planning
 - ✓ Non-medical related financial assistance
 - ✓ Information and community referral

6. Financial Counsellor

- Conduct financial assessment and administer financial assistance for patients who face difficulties with their medical bills.

REFERRALS

- Refer to Sleep Unit (under ENT or Respiratory subspecialty) if you suspect medical issues such as obstructive sleep apnoea, restless legs syndrome.
- Consider pointing patients to family service centres (FSC) for supportive counselling for those with social issues. Search online using “FSC locator” and key in patient’s postal code for the nearest FSC available.
- TCU Medical Social Worker if patient requires brief supportive counselling especially for patients with complex psychosocial setups, or for subthreshold insomnia.
- **TCU Psychologist Counselling (Short)(First Visit FV) for non-pharmacological interventions; this should be first line in management of primary insomnia.**
- TCU Dr Health and Mind (Long) FV if you suspect that patient may have underlying depression or anxiety that require longitudinal follow up, or if patients require medications. You may trial antihistamines or SSRIs depending on the primary pathology.
- Refer to NUHS Psychological Medicine department (depending on patient’s home address) if patient has failed trial of at least 2 agents at adequate dose and duration or require prolonged use of benzodiazepines or Z-drugs.

ABBREVIATIONS

BPSD	Behavioural and Psychological Symptoms of Dementia
BZD	Benzodiazepines
CBT	Cognitive Behavioural Therapy
DSWPD	Delayed Sleep Wake Phase Disorder
FSC	Family Service Centre
OTC	Over the counter

REFERENCES

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